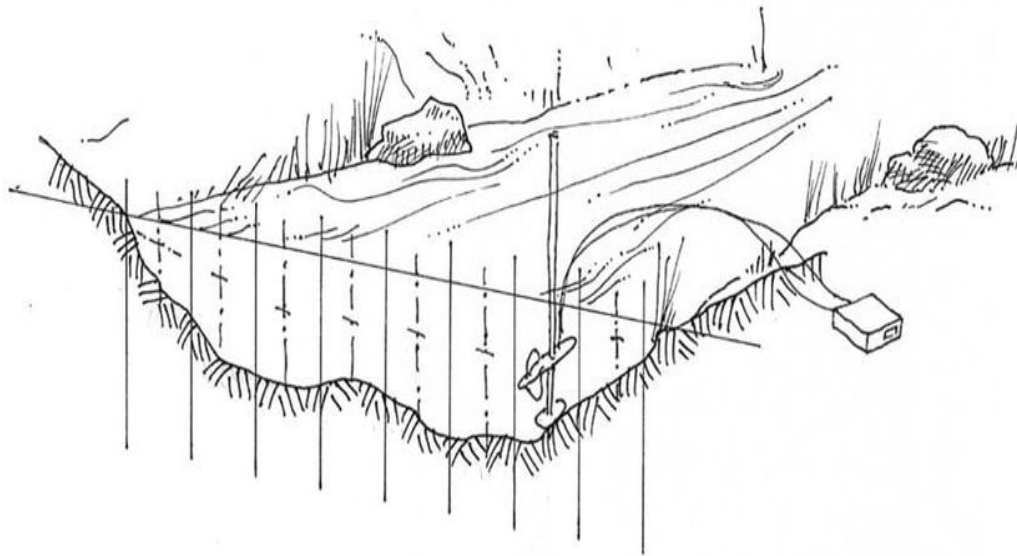


1. **Describe the standard method of computing discharge in a stream after measurement of velocity at various verticals by current meter.**

Answer:

In this method, the stream cross-section is divided into several vertical sections, the velocity at each vertical is measured with a current meter. The discharge is computed by the velocity-area method.

The standard method of computing discharge in a stream after measurement of velocity at various verticals by current meter is given below:



- i. A straight, uniform reach of the stream is selected.
- ii. The flow must be free from eddies, bends, weeds and obstructions.
- iii. Stream bed should be stable.
- iv. Cross-section of the river is divided into vertical sections depending upon the stream width.
- v. The depth of water is measured at each vertical using sounding weight.
- vi. Velocity of the flow is measured at each vertical by using current meter.
- vii. Mean velocity of the flow is determined as follows:

Depth less than 0.75 m: Measure velocity at 0.6D below the water surface.

➤ Mean velocity = Velocity at 0.6D.

Depth greater than 0.75 m: Measure velocities at 0.2D and 0.8D below the water surface.

➤ Mean velocity:

$$V = \frac{V_{0.2} + V_{0.8}}{2}$$

- viii. Area of each vertical strip is computed as:

$$A_i = b_i \times d_i$$

where :

$$A_i = \text{Area of strip (m}^2\text{)}$$

$b_i = \text{Width of strip (m)}$

$d_i = \text{Average depth (m)}$

ix. Discharge of each strip is computed as:

$$Q_i = A_i \times V_i$$

where:

$Q_i = \text{Discharge of strip (m}^3/\text{s)}$

$V_i = \text{Mean velocity of the strip (m/s)}$

x. Total discharge is computed as:

$$Q = \sum_{i=1}^n A_i V_i$$

where:

$Q = \text{Total stream discharge (m}^3/\text{s)}$

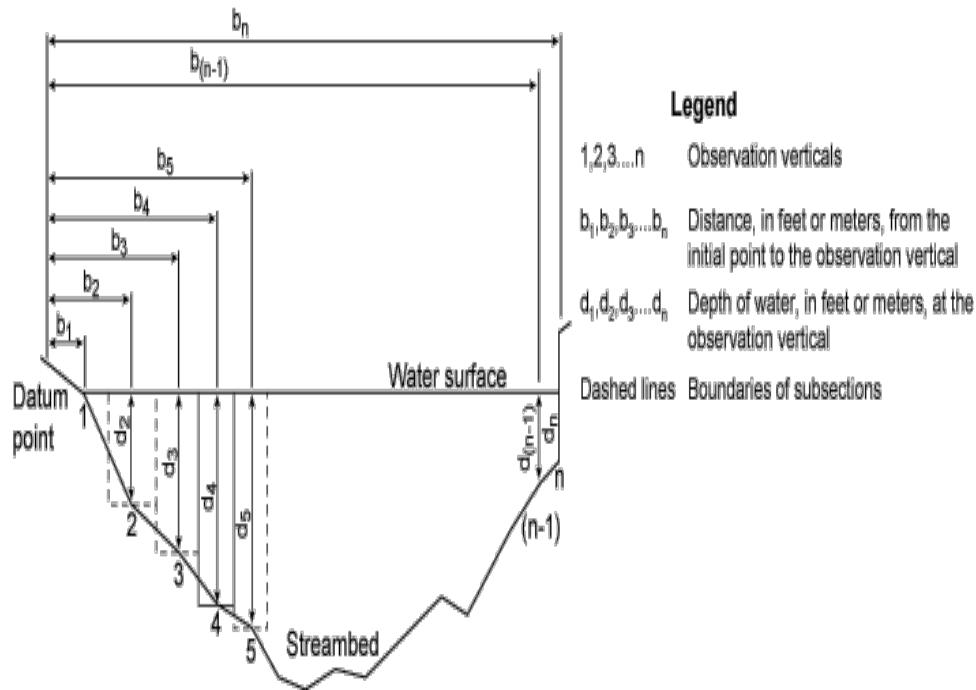
2. **What are the data to be obtained from the field measuring to determine the discharge by slope area and current meter methods? Explain how the discharge is obtained?**

Answer:

Slope-area method is an indirect technique to compute peak flood discharge in a river using high-water marks, channel geometry, and Manning's roughness equation.

The data to be obtained from the field measuring to determine the discharge by slope area method are stated below:

- i. Shape of the stream
- ii. Width of stream
- iii. Depth of water at several points
- iv. Wetted perimeter
- v. Water surface slope and bed slope
- vi. Length of selected reach of river
- vii. Type of channel surface



Procedure:

- i. Selection of suitable river reach:
 - A straight, stable and uniform reach of river is chosen.
 - The bends, confluences, bridges and backwater effects are avoided.
- ii. Conduction of cross-section survey:
 - At least two cross-sections are surveyed.
 - Flow area and wetter perimeter is measured.
- iii. Identification of high flood marks:
 - The flood marks are located on both river banks.
 - Flood water surface elevations are determined.
- iv. Measurement of reach length:
 - The distance between the selected cross-sections along the river is to be determined.
- v. Calculation of water surface slope:
 - The difference in flood water levels between the two sections is determined.
 - Slope is computed as:

$$S = \frac{\Delta h}{L}$$

where :

S = Water surface slope

Δh = Difference $\in$ water level

L = Length of river reach

- vi. Calculation of hydraulic radius:
 - Compute:

$$R = \frac{A}{P}$$

where :

$$A = \text{Cross-sectional area (m}^2\text{)}$$

$$P = \text{Wetted perimeter (m)}$$

vii. Selection of Manning's coefficient:

- The appropriate value of Manning's roughness coefficient is estimated based on the river bed material, vegetation etc.

viii. Calculation of flow velocity:

- Flow velocity is computed by using the formula:

$$V = \frac{1}{n} R^{2/3} S^{1/2}$$

ix. Computation of river discharge:

- The river discharge was determined using:

$$Q = AV$$

where :

$$Q = \text{Discharge (m}^3/\text{s)}$$

$$A = \text{Flow area (m}^2\text{)}$$

$$V = \text{Mean velocity (m/s)}$$

The data to be obtained from the field measuring to determine the discharge by current meter methods are listed as follows:

- i. Width of stream
- ii. Spacing of verticals across the stream
- iii. Water depth at each vertical
- iv. Velocity at each vertical measured by current meter
- v. Cross-sectional area of each strip.

Procedure:

- a. A straight, uniform reach of the stream is selected.
- b. The flow must be free from eddies, bends, weeds and obstructions.
- c. Stream bed should be stable.
- d. Cross-section of the river is divided into vertical sections depending upon the stream width.
- e. The depth of water is measured at each vertical using sounding weight.
- f. Velocity of the flow is measured at each vertical by using current meter.
- g. Mean velocity of the flow is determined as follows:
 - Depth less than 0.75 m:** Measure velocity at 0.6D below the water surface.
 - Mean velocity = Velocity at 0.6D.
 - Depth greater than 0.75 m:** Measure velocities at 0.2D and 0.8D below the water surface.

$$\text{Mean velocity: } V = \frac{V_{0.2} + V_{0.8}}{2}$$

- h. Area of each vertical strip is computed as:

$$A_i = b_i \times d_i$$

where:

$$A_i = \text{Area of strip (m}^2\text{)}$$

$$b_i = \text{Width of strip (m)}$$

$$d_i = \text{Average depth (m)}$$

- i. Discharge of each strip is computed as:

$$Q_i = A_i \times V_i$$

where:

$$Q_i = \text{Discharge of strip (m}^3/\text{s)}$$

$$V_i = \text{Mean velocity of the strip (m/s)}$$

- j. Total discharge is computed as:

$$Q = \sum_{i=1}^n A_i V_i$$

where:

$$Q = \text{Total stream discharge (m}^3/\text{s)}$$

3. a. A watershed has five rain gauges, the annual rainfall recorded by those gauges for 2007 are as follows:

Rain gauges	1	2	3	4	5
Annual rainfall (cm)	88	106	150	80	100

Calculate the optimum number of rain gauges for the watershed if a 10% error in the mean areal rainfall is acceptable.

- b. Explain the risk of GLOFs?

Answer:

- a.

Optimum number of rain gauges is determined by the formula:

$$N = \left(\frac{C_v}{E} \right)^2$$

where:

$$N = \text{Optimum number of rain gauges}$$

$$C_v = \text{Coefficient of variation of rainfall (\%)}$$

$$E = \text{Allowable percentage error} = 10\%$$

Mean rainfall is determined by:

$$\bar{P} = \frac{88 + 106 + 150 + 80 + 100}{5} = \frac{524}{5} = 104.8 \text{ cm}$$

Standard deviation of rainfall is computed:

$$S = \sqrt{\frac{\sum (P - \bar{P})^2}{n-1}} = \sqrt{\frac{2964.8}{4}} = \sqrt{741.2} = 27.23 \text{ cm}$$

Calculate coefficient of variation:

$$C_v = \frac{S}{\bar{P}} \times 100 = \frac{27.23}{104.8} \times 100 = 25.98\%$$

$$N = \left(\frac{25.98}{10} \right)^2 = (2.598)^2 = 6.75$$

Hence ,

$$N \approx 7 \text{ rain gauges}$$

Since 5 rain gauges already exist, the watershed requires: $7-5=2$ additional rain gauges.

b.

A Glacial Lake Outburst Flood (GLOF) is the sudden release of a large volume of water from a glacial lake due to the failure of its natural dam such as a moraine or ice dam. GLOFs are a major natural hazard in the Himalayan region, including Nepal. Some of the well-known potentially dangerous glacial lakes includes Tsho Rolpa Glacial Lake and Imja Tsho. Furthermore, the melting of glaciers at increased rates creates the risk of Glacial Lake Outburst Floods (GLOFs), which can destroy downstream ecosystems and structures.

The risk of GLOFs can be described as follows:

- i. Sudden floods may cause deaths and injuries in downstream settlements due to little warning time.
- ii. Infrastructures such as roads, bridges, hydropower plants, irrigation canals may be destroyed by GLOFs.
- iii. Floodwaters erode the fertile land, destroys crops and livestock affecting livelihood of farmers.
- iv. GLOFs damages the forests, eco-systems, wildlife habitats and reduces the water quality by carrying debris and sediments.
- v. Climate changes accelerates the glacier melting which increases the potential future hazard.

4. **a. 60 mm of runoff depth is obtained from a catchment of area 2000 km². Calculate the runoff volume in million cubic meters, cumec-day and hectare-metres.**
b. Briefly describe how a flow duration curve is constructed.
c. Give three important application of flow duration curve.

Answer:

Given,

$$\text{Runoff depth, } d = 60 \text{ mm} = 0.06 \text{ m}$$

$$\text{Catchment area, } A = 2000 \text{ km}^2 = 2000 \times 10^6 \text{ m}^2$$

Calculation of Runoff volume:

$$V = A \times d = (2000 \times 10^6) \times 0.06 = 120 \times 10^6 \text{ m}^3$$

$$\text{Runoff volume} = 120 \text{ million m}^3$$

We know,

1 cumec-day = discharge of 1 m³/s flowing for 1 day.

$$1 \text{ cumec - day} = 86400 \text{ m}^3$$

$$\text{Cumec - days} = \frac{120 \times 10^6}{86400}$$

$$= 1388.89 \text{ cumec-days}$$

$$\text{Runoff} = 1388.9 \text{ cumec-days}$$

$$1 \text{ hectare} = 10,000 \text{ m}^2$$

$$1 \text{ hectare - metre} = 10,000 \text{ m}^3$$

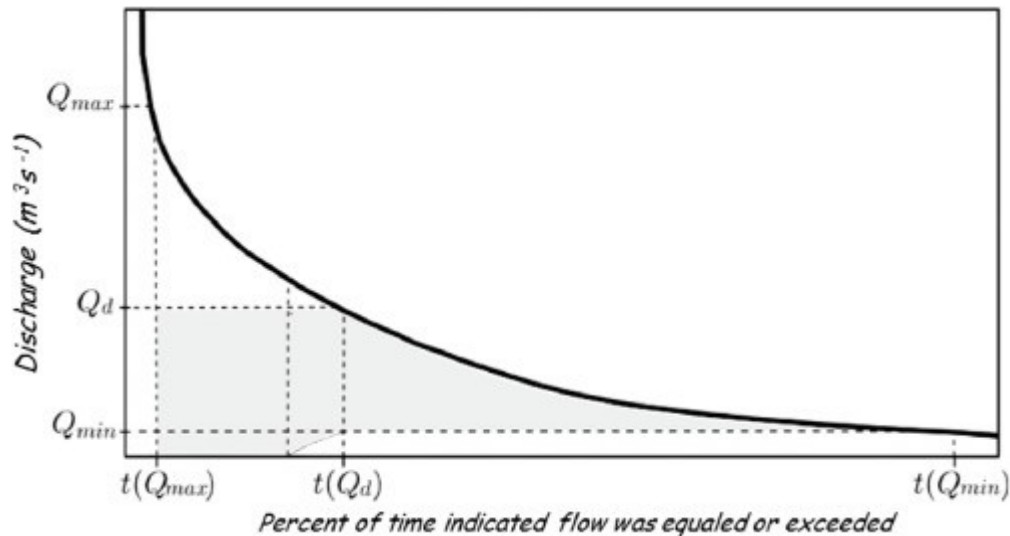
$$\text{Hectare-metres} = \frac{120 \times 10^6}{10000}$$

$$\text{Runoff} = 12,000 \text{ hectare-metres}$$

b. A Flow Duration Curve (FDC) is a plot showing the percentage of time that a given discharge is equaled or exceeded. It is widely used to determine the design discharge and hence the installed capacity of a Run-of-River hydropower plant. It is constructed by following procedure:

Procedure:

1. Collect long-term streamflow data.
2. Arrange discharges in descending order.
3. Compute exceedance probability.
4. Plot the Flow Duration Curve.
5. Select the design discharge (e.g., Q_{40} , Q_{50} , or Q_{60}).
6. Calculate installed capacity using: $P = \rho g Q H \eta$.



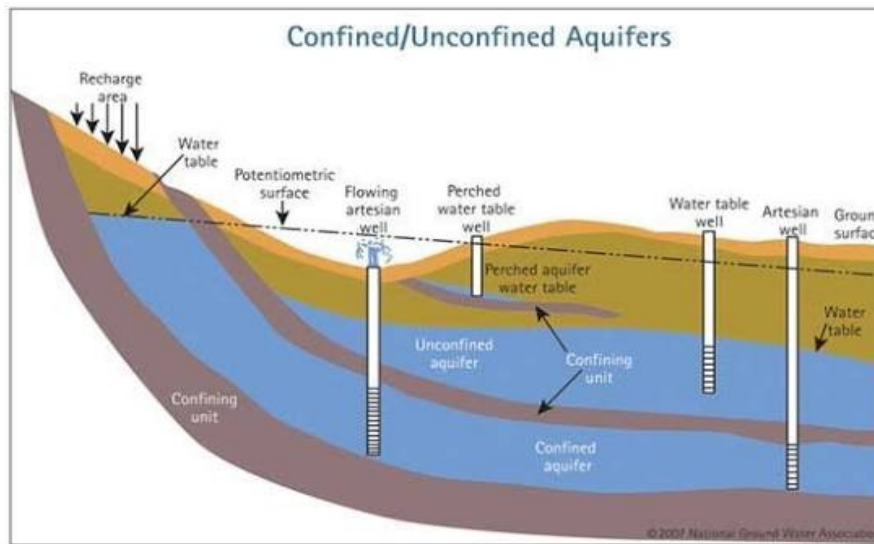
c. The three important application of flow duration curve is stated below:

- i. Hydropower planning and design:
Flow Duration Curve is used to estimate the amount of water available for power generation and selection of the installed capacity of hydropower plants.
- ii. Water resources planning and design:
Flow Duration Curve also helps to assess the reliability of streamflow for irrigation, municipal water supply and estimate the storage capacity of reservoirs.
- iii. Assessment of stream flow characteristics:
Flow duration curve is also useful for analyzing the flows of river for river management, environmental flow studies and drought analysis.

5. **Derive an expression for the steady radial flow to a well in an unconfined aquifer.**

Answer:

Unconfined aquifer is the type of aquifer in which water table forms the upper surface of zone of saturation. It is also known as phreatic aquifer or non-artesian aquifer.



The derivation for the expression for the steady radial flow to a well in an unconfined aquifer is given below:

Hydraulic conductivity = K

Radius of influence = R

Radius of well = r_w

Water table height at radius $r = h$

Water table height at radius $R = H$

Water table height in the well = h_w

Discharge through a cylindrical surface of radius r :

$$Q = K \times A \times i$$

where ,

$$\text{Flow area, } A = 2\pi rh$$

$$\text{Hydraulic gradient, } i = \frac{dh}{dr}$$

Since the head decreases $\rightarrow$ the well ,

$$Q = -K(2\pi rh) \frac{dh}{dr}$$

or

$$Q = -2\pi K r h \frac{dh}{dr}$$

$$Q dr = -2\pi K r h dh$$

Divide both sides by $2\pi K$:

$$\frac{Q}{2\pi K} \frac{dr}{r} = -h dh$$

$$\text{At } r = r_w, h = h_w$$

At $r = R, h = H$

$$\frac{Q}{2\pi K} \int_{r_w}^R \frac{dr}{r} = - \int_{h_w}^H h dh$$

$$\frac{Q}{2\pi K} \ln\left(\frac{R}{r_w}\right) = \frac{H^2 - h_w^2}{2}$$

$$Q = \frac{\pi K (H^2 - h_w^2)}{\ln\left(\frac{R}{r_w}\right)}$$

This is the Dupit's equation for steady radial flow in an unconfined aquifer.

6. **The annual peak discharge of a river follows the Gumbel's extremes value distribution with a mean of 10000 m³/s and a standard deviation of 3000 m³. What is the probability that the annual peak discharge is more than 15000 m³/s. What is the magnitude of the peak discharge with an exceedance probability of 0.1?**

Answer:

Given,

Mean annual peak discharge, $\bar{X} = 10000 \text{ m}^3/\text{s}$

Standard deviation, $S = 3000 \text{ m}^3/\text{s}$

For Gumbel extreme value distribution,

$$\alpha = \frac{\pi}{\sqrt{6} S} = \frac{1.2825}{S}$$

$$u = \bar{X} - 0.5772 \left(\frac{1}{\alpha} \right)$$

$$\alpha = \frac{1.2825}{3000} = 0.0004275$$

$$\frac{1}{\alpha} = 2339.18$$

$$u = 10000 - 0.5772 (2339.18)$$

$$u = 10000 - 1350.58$$

$$u = 8649.42 \text{ m}^3/\text{s}$$

- a. Probability that annual peak discharge exceeds 15000 m³/s

For Gumbel distribution,

$$F(x) = e^{-e^{-\alpha(x-u)}}$$

where $F(x)$ is the probability that discharge is less than or equal to x .

$$y = \alpha(x - u)$$

$$= 0.0004275 (15000 - 8649.42)$$

$$= 2.715$$

$$F(15000) = e^{-e^{-2.715}}$$

We know,

$$e^{-2.715} = 0.0663$$

$$F = e^{-0.0663} = 0.9358$$

$$P(X > 15000) = 1 - F$$

$$= 1 - 0.9358$$

$$P(X > 15000) = 0.064 \approx 6.4 \%$$

$$P(X > 15000) = 0.064 \approx 6.4 \%$$

b. Peak discharge with exceedance probability=0.1

$$\text{Exceedance probability, } P(X > x) = 0.1$$

Therefore,

$$F(x) = 0.9$$

$$\text{Reduced variate, } y = -\ln[-\ln(F)]$$

$$= -\ln[-\ln(0.9)]$$

$$= 2.250$$

$$x = u + \frac{y}{\alpha}$$

$$x = 8649.42 + 2.2500.0004275$$

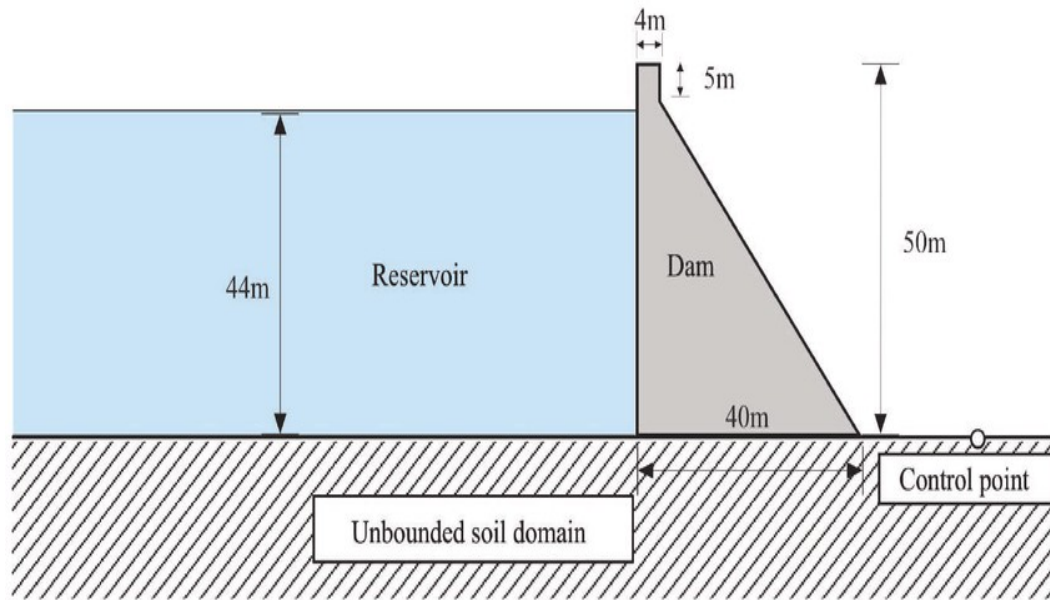
$$= 8649.42 + 5263.16$$

$$x = 13913 \text{ m}^3/\text{s}$$

7. **How do you fix the height of dam for the particular site? What do you understand by trap efficiency of the reservoir?**

Answer:

The height of a dam is determined by considering the storage requirements, site conditions, safety, and economic feasibility. It should provide the required reservoir capacity while ensuring structural stability and safe flood passage.



The height of dam for the particular site is fixed by considering the following points:

- i. The dam should provide sufficient storage for irrigation, hydropower, water supply and other purposes.
- ii. The shape and elevation of the valley also plays an important role in determining the storage obtained for a given height of the dam.
- iii. The height is fixed after determining full reservoir level, maximum water level, design flood and so on.
- iv. Additional height is provided in the dam above the maximum water level to prevent overtopping due to wave action, wind, floods etc.
- v. The foundation must be strong and stable enough to support the dam safely.
- vi. Adequate storage must be reserved for sediment accumulation during the design life of reservoir.

The formula for determining dam height is:

$$\text{Height of Dam} = \text{Crest Level} - \text{Lowest Foundation Level}$$

$$\text{Crest Level} = \text{MWL} + \text{Freeboard}$$

Trap efficiency is the percentage of the incoming sediment that is retained in a reservoir instead of passing through it downstream. It indicates how effectively a reservoir stores sediment.

$$\text{Trap Efficiency (TE)} = \frac{\text{Sediment trapped}}{\text{Total sediment inflow}} \times 100 \%$$

$$\text{Trap Efficiency} = \frac{\text{Total sediment inflow} - \text{Sediment trapped}}{\text{Total sediment inflow}} \times 100 \%$$

or

$$\text{TE} = \frac{S_i - S_o}{S_i} \times 100 \%$$

where :

$$S_i = \text{Sediment inflow}$$

$$S_o = \text{Sediment outflow}$$

Factors affecting trap efficiency:

- i. Reservoir capacity-to-inflow ratio:
Larger reservoirs generally have higher trap efficiency.
- ii. Particle size of sediment:
Coarse particles settle more easily than fine particles.
- iii. Flow velocity:
Lower velocities promote sediment settling.
- iv. Reservoir shape and depth:
Deep and wide reservoirs trap more sediment.
- v. Mode of reservoir operation:
Frequent drawdown may reduce trap efficiency.

Advantages of trap efficiency:

- i. It helps to estimate the rate of reservoir sedimentation.
- ii. It determines the useful life of a reservoir.
- iii. It is important for the design of dams, spillways and outlet works.

8. **Describe two techniques to measure suspended-sediment discharge. How do they differ in the evaluation of suspended sediment concentration?**

Answer:

Suspended sediment discharge of a river is determined by measuring sediment concentration in the flowing water and multiplying it by water discharge. The two techniques to measure suspended sediment discharge are:

i. Point sampling techniques:

- A sediment sampler is placed at a single point in the flow, usually at 0.6 depth or another selected depth.
- A water-sediment sample is collected at that point.
- The sample is analyzed in the laboratory to determine the suspended sediment concentration.
- Sediment discharge is calculated as:

$$Q_s = Q \times C$$

where:

Q_s = suspended sediment discharge

Q = water discharge

C = suspended sediment concentration

ii. Depth-integrated sampling techniques:

- A depth-integrating sampler is moved uniformly from the water surface to the bed and back at each vertical.
- Samples are collected from several verticals across the stream.
- The combined samples represent the average suspended sediment concentration over the entire cross-section.
- Suspended sediment discharge is then computed using the average concentration and stream discharge.

The difference between the two techniques is listed below:

Point sampling	Depth-integrated sampling
Sediment Concentration is measured at one point in vertical.	Sediment concentration is averaged over the entire depth at each vertical.
It assumes the sampled concentration represents the whole flow.	It accounts for variation of concentration with depth.
Point sampling has lower accuracy.	Depth-integrated sampling has higher accuracy.
The technique is faster and simpler.	The technique is more time-consuming and reliable.

9. a. Describe with neat sketch how the storage volume of reservoir for the given rate of draft is fixed through mass curve analysis.

b. Calculate the suspended sediment discharge (in kilonewtons per day) for a suspended sediment concentration of 22,000 ppm and a flow of 155 m³/s.

Answer:

a.

A mass inflow curve is a graph showing the cumulative inflow volume into a reservoir plotted against time. It is used to determine the storage (reservoir) capacity required to meet a constant water demand.

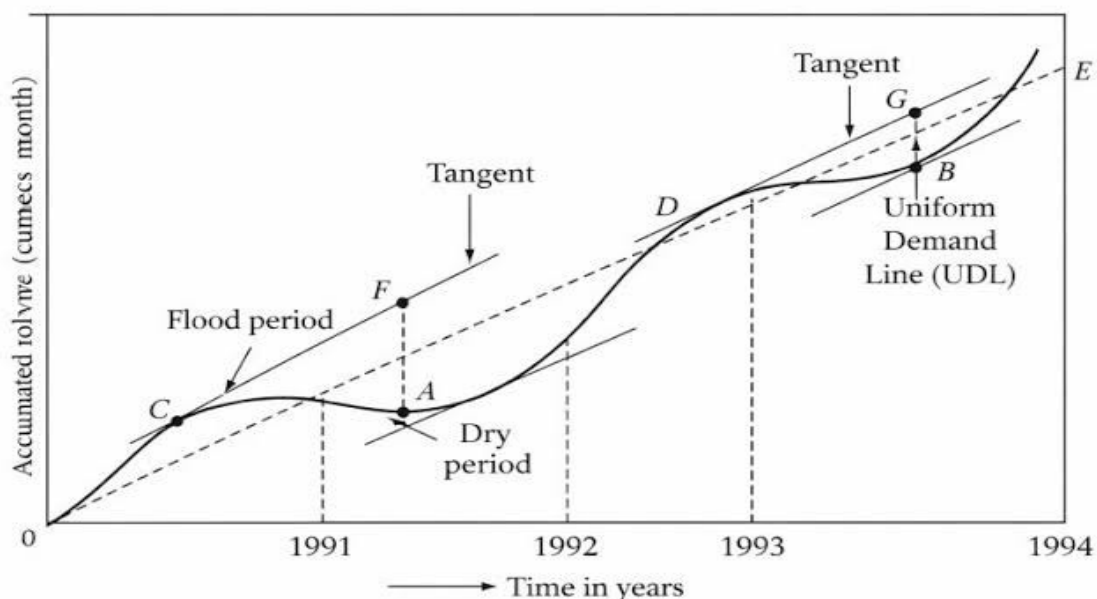
Principle:

- If the cumulative inflow is greater than the cumulative demand, water is stored.
- If the cumulative demand exceeds cumulative inflow, water is supplied from storage.
- The maximum difference between the cumulative demand line and the mass inflow curve gives the required reservoir capacity.

Procedure:

- a. The monthly or daily streamflow data is collected.
- b. The cumulative inflow volume is computed.
- c. The cumulative inflow (Y-axis) versus time (X-axis) is plotted.
- d. A straight demand line with the slope equals to constant demand is drawn.
- e. Then, the parallel demand lines touching the mass curve is also drawn.
- f. The maximum vertical distances between the demand line and the mass curve is measured.

Hence, Reservoir Capacity = Maximum cumulative deficit



b.

Given,

Suspended sediment concentration, $C = 22,000 \text{ ppm}$

Water discharge, $Q = 155 \text{ m}^3/\text{s}$

For water: $1 \text{ ppm} \approx 1 \text{ mg/L}$

Therefore,

$$C = 22,000 \text{ mg/L} = 22 \text{ kg/m}^3$$

$$Q_s = C \times Q$$

$$Q_s = 22 \times 155 = 3410 \text{ kg/s}$$

$$3410 \times 86400 = 294,624,000 \text{ kg/day}$$

We know,

$$W = m \times g$$

$$\text{where } g = 9.81 \text{ m/s}^2$$

$$W = \frac{294,624,000 \times 9.81}{1000}$$

$$W \approx 2.89 \times 10^6 \text{ kN/day}$$

10. what do you understand by hydrological forecasts and warnings? Describe the classification of hydrological forecasts and types of warnings.

Answer:

The prediction of future hydrological events or conditions, such as river discharge, water level, floods, droughts, reservoir inflows, and runoff, using hydrological and meteorological data is known as hydrological forecasts. It helps in water resources management and disaster preparedness.

Similarly, a hydrological warning is an official notice issued when hydrological event such as flood, landslide is occurring so that people can take protective measures before the catastrophe.

Hydrological forecasts can be classified as follows:

i. Short term forecast:

- If the forecast period is of few hours to 2-3 days, it is known as short-term forecast.
- It is used for flood forecasting, reservoir operation and emergency response.

ii. Medium-term forecast:

- If the period of forecast is from 3 to 10 days, it is known as medium-term forecast.
- It is used for reservoir management, water supply planning etc.

iii. Long-term forecast:

- If the period of forecast is from several weeks to months, it is known as medium-term forecast.
- It is used for drought assessment, hydropower generation and long-term water resource management.

Hydrological warnings are classified into:

i. Flood watch:

- It indicates that the conditions are pointing towards flood occurrence.
- People should remain alert and monitor updates.

- ii. Flood warning:
 - Flooding is expected to occur or is already occurring.
 - Immediate preparedness and protective actions are required.
- iii. Drought warning:
 - It is issued when prolonged rainfall deficiency leads to water shortages affecting agriculture, water supply and environment.